

SUMMARY OF ADVANTAGES AND INDUSTRIAL APPLICABILITY (NON-LIMITING)

A. Overview

In various embodiments, the disclosed system provides a multi-node architecture for collecting intimate physiology signals and geometry-related signals over time, deriving privacy-preserving metrics, and enabling cross-device correlation and cross-user correlation under consent gating. The embodiments are configured to improve interpretability by combining intimate-region signals with upper-body context and by emphasizing longitudinal patterns rather than isolated snapshots.

B. Example Advantages (Non-Limiting)

In certain embodiments, the system provides one or more of the following advantages: 1. Longitudinal measurement of intimate-region signals during sleep and daily life, enabling trends and stability measures rather than momentary snapshots. 2. Context-conditioned interpretation using upper-body physiologic context, improving robustness and reducing misinterpretation from motion artifacts, temperature shifts, and contact instability. 3. Baseline establishment and normalization using geometry and/or compliance workflows, improving repeatability and cross-session comparability. 4. Multi-modal sensing flexibility allowing sensor substitution while retaining functional outputs through privacy-preserving metrics and confidence labeling. 5. Privacy-preserving presentation using indices, zones, ranges, curves, and confidence indicators without explicit anatomical imagery or explicit anatomical labeling. 6. Cross-device correlation enabling multi-node interpretation, including time alignment, context-tagged segmentation, and correlation scoring. 7. Cross-user correlation under consent enabling alignment, synchrony, and stability scoring using derived metrics rather than raw signal sharing, supporting diverse pairing contexts.

C. Industrial Applicability

In various embodiments, the disclosed system may be used in consumer and/or clinical contexts, including wellness tracking, longitudinal pattern monitoring, recovery tracking, and other physiologic pattern monitoring contexts. The system may be implemented using wearable electronics, embedded sensor devices, mobile computing platforms, and/or server-based computing infrastructure. The system may be manufactured using medical-grade materials, consumer-grade wearable materials, or combinations thereof, and may be deployed at scale through mobile applications and device distribution channels.

D. Non-Limiting Closing Statement

The described embodiments are illustrative only. The scope of the disclosure includes all variations, substitutions, and equivalents consistent with the described systems and methods, including any combination of nodes, sensor modalities, processing modules, correlation methods, privacy-preserving outputs, and workflow sequences.